MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M752A1
PARTS FOR

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for the fabrication of parts and packing for M752A1, Ignition Cartridges for 81MM, M853 Illuminating and M819 Smoke Cartridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form part of the specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9294752 - Cartridge, Ignition M752A1
9294754 - Tube
9294756 - Cap
9294758 - Tube, Flash
9294755 - Body
9293427 - Head
9243255 - Paper, Cartridge
12630552 - Plug, Firing

(Copies of specifications, standards, drawings and publications listed above or in the item detail specification required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing (dwg.) 9294752 and associated drawings. and with all requirements specified in applicable specifications and standards.

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3.3 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Bursting strength. The tube assemblies shall not burst at a compression load of 57 pounds, minimum, but shall burst at a compression load of 192 pounds, maximum. Testing shall be as specified in 4.4.3.1.

3.5 Workmanship. All parts shall be fabricated and furnished in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawing	Quantity
Plug, firing	12630552	25
Head	9293427	25
Tube	9294754	25
Body	9294755	25
Cap	9294756	25
Tube, flash	9294758	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 (AR).

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Plug, firing (dwg. 12630552)				
	Examination for defects	25		3.2	4.4.2.3
	Head (dwg. 9293427)				
	Examination for defects	25		3.2	4.4.2.4
	Tube (dwg. 9294754)				
	Examination for defects	25		3.2	4.4.2.1
	Burst strength	25(a)		3.2	4.4.3.1
	Body (dwg. 9294755)				
	Examination for defects	25		3.2	4.4.2.2
	Cap (dwg. 9294756)				
	Examination for defects	25		3.2	4.4.2.6
	Tube, flash (dwg. 9294758)				
	Examination for defects	25		3.2	4.4.2.5

NOTES:
(a) The same samples submitted for examination shall be used for the burst test.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9294754
4.4.2.1	Tube				NEXT HIGHER ASSEMBLY 9294753
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Length		0.40%	3.2	Gage
103	Inside diameter		0.40%	3.2	Gage
104	Tube not spiral wound	5		3.2	Visual
105	Burst strength		0.65%	3.4	Gage/4.4.3.1
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body	1 OF 2	9294755
			NEXT HIGHER ASSEMBLY 9294752

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash hole missing or obstructed		100%	3.2	Visual
102	Pitch diameter of threads, min.		0.40%	3.2	Gage
103	Major diameter of threads, min.		0.40%	3.2	Gage
104	Diameter of primer cavity		0.40%	3.2	Gage
105	Depth of primer cavity		0.40%	3.2	Gage
106	Depth of flash tube cavity		0.40%	3.2	Gage
107	Diameter of flash tube cavity		0.40%	3.2	Gage
108	Small diameter of flash hole		0.40%	3.2	Gage
109	Large diameter of flash hole		0.40%	3.2	Gage
110	Large outside diameter		0.40%	3.2	Gage
111	Diameter of pellet cavity		0.40%	3.2	Gage
112	Depth to bottom of pellet cavity		0.40%	3.2	Gage
113	Length overall		0.40%	3.2	Gage
114	True position of primer cavity		0.40%	3.2	Gage
115	Width of groove		0.40%	3.2	Gage
116	Depth of groove		0.40%	3.2	Gage
117	Small outside diameter		0.40%	3.2	Gage
118	Diameter of countersink at base of primer cavity, max.		0.40%	3.2	Gage
119	Length from rear face to large outside diameter		0.40%	3.2	Gage
120	Outside diameter forward and rear of groove		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9294755
4.4.2.2	Body (Cont)				NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
121	Length from large diameter to end of external thread		0.40%	3.2	Gage
122	Length from large diameter to beginning of thread		0.40%	3.2	Gage
123	Diameter of thread undercut		0.40%	3.2	Gage
124	Length of large diameter section		0.40%	3.2	Gage
125	Length from front face to groove		0.40%	3.2	Gage
126	Depth from forward face to flash hole cavity rear of pellet cavity		0.40%	3.2	Gage
127	Protective finish missing		0.40%	3.2	Visual
128	True position of forward outside diameter with P.D. of thread		0.40%	3.2	Gage
Minor					
201	Radii or chamfers missing or incorrect		0.65%	3.2	Visual/6.4
202	Surface finish improper		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Plug, Firing	1 OF 1	9293426
			NEXT HIGHER ASSEMBLY: 9294752/9391019

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of tip		0.40%	3.2	Gage
102	Total length including tip		0.40%	3.2	Gage
103	Length from tip to small diameter		0.40%	3.2	Gage
104	Radius improper		0.40%	3.2	Gage
105	Large diameter		0.40%	3.2	Gage
106	Small diameter		0.40%	3.2	Gage
107	Protective finish missing		0.40%	3.2	Visual
108	Chamfer missing or incorrect		0.40%	3.2	Visual/6.4
109	True position of tip with large outside diameter		0.40%	3.2	Gage
Minor					
201	Surface finish improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9293427
4.4.2.4	Head				NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of internal threads		0.40%	3.2	Gage
102	Minor diameter of internal threads		0.40%	3.2	Gage
103	Pitch diameter of external threads		0.40%	3.2	Gage
104	Major diameter of external threads		0.40%	3.2	Gage
105	Length of internal thread, min.		0.40%	3.2	Gage
106	Overall length		0.40%	3.2	Gage
107	Length of large internal diameter		0.40%	3.2	Gage
108	Diameter of firing plug hole		0.40%	3.2	Gage
109	Large outside diameter		0.40%	3.2	Gage
110	Small outside diameter		0.40%	3.2	Gage
111	Large inside diameter		0.40%	3.2	Gage
112	True position of large inside diameter with P.D. of external thread		0.40%	3.2	Gage
113	True position of firing plug hole with P.D. of external thread		0.40%	3.2	Gage
114	True position of internal threads with external threads		0.40%	3.2	Gage
115	Length from shoulder to end of external threads		0.40%	3.2	Gage
116	Protective finish missing		0.40%	3.2	Visual
Minor					
201	Surface finish improper		0.65%	3.2	Visual
202	Diameter of spanner holes		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9293427
4.4.2.4	Head (Cont)				NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Depth of spanner holes		0.65%	3.2	Gage
204	Location of spanner holes		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.5	Visual
206	Distance from front face to shoulder		0.65%	3.2	Gage
207	Length of thread undercut		0.65%	3.2	Gage
208	Chamfer missing or incorrect		0.65%	3.2	Visual/6.4

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Tube, Flash				9294758
					NEXT HIGHER ASSEMBLY 9294757
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Longitudinal flash hole missing or obstructed		100%	3.2	Visual
102	Flash hole(s) missing or excessive in number		100%	3.2	Visual
103	True position of flash holes		0.40%	3.2	Gage
104	Diameter of flash holes		0.40%	3.2	Gage
105	Overall length		0.40%	3.2	Gage
106	Large outside diameter		0.40%	3.2	Gage
107	Small outside diameter		0.40%	3.2	Gage
108	Length of small outside diameter		0.40%	3.2	Gage
109	Large inside diameter		0.40%	3.2	Gage
110	Length of large inside diameter		0.40%	3.2	Gage
111	Protective finish inadequate		0.40%	3.2	Visual
112	Small inside diameter		0.40%	3.2	Gage
113	Distance from rear face to closest flash hole		0.40%	3.2	Gage
Minor					
201	Surface finish improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48396A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cap	1 OF 1	9294756
			NEXT HIGHER ASSEMBLY 9294752

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Total length		0.40%	3.2	Gage
103	Depth of groove		0.40%	3.2	Gage
104	Width of groove		0.40%	3.2	Gage
105	Diameter of cavity		0.40%	3.2	Gage
106	Depth of cavity		0.40%	3.2	Gage
107	Length from edge to beginning of groove		0.40%	3.2	Gage
108	Protective finish inadequate		0.40%	3.2	Visual
109	True position of cavity with outside diameter		0.40%	3.2	Gage
Minor					
201	Radii or chamfers missing or improper		0.65%	3.2	Visual/6.4
202	Surface finish improper		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing

4.4.3.1 Bursting strength (see 3.4) Major Defect. See Figure I.

4.4.3.1.1 Sampling plan at the beginning of production. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-105 Code Letter J and AQL of 0.65 percent. The test shall be performed as specified in 4.5.1.

4.4.3.1.2 Sampling plan after three lots of production. After three consecutive lots have met the criteria of 4.4.3.1.1, the sampling plan for this test shall be in accordance with MIL-STD-105 Code Letter F, using an AQL of 0.65%. The test shall be performed as specified in 4.5.1.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Methods column. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Bursting strength. The paper tube (without heat shrinkable tubing) shall be cut in half lengthwise and one of the halves placed in the fixture (See Figure 1). The stop (3) shall be completely removed or backed off so that it does not touch the punch (2). Place fixture in universal tester set for a travel speed of 0.125 inches per minute. Proceed with Test Number 1 and 2.

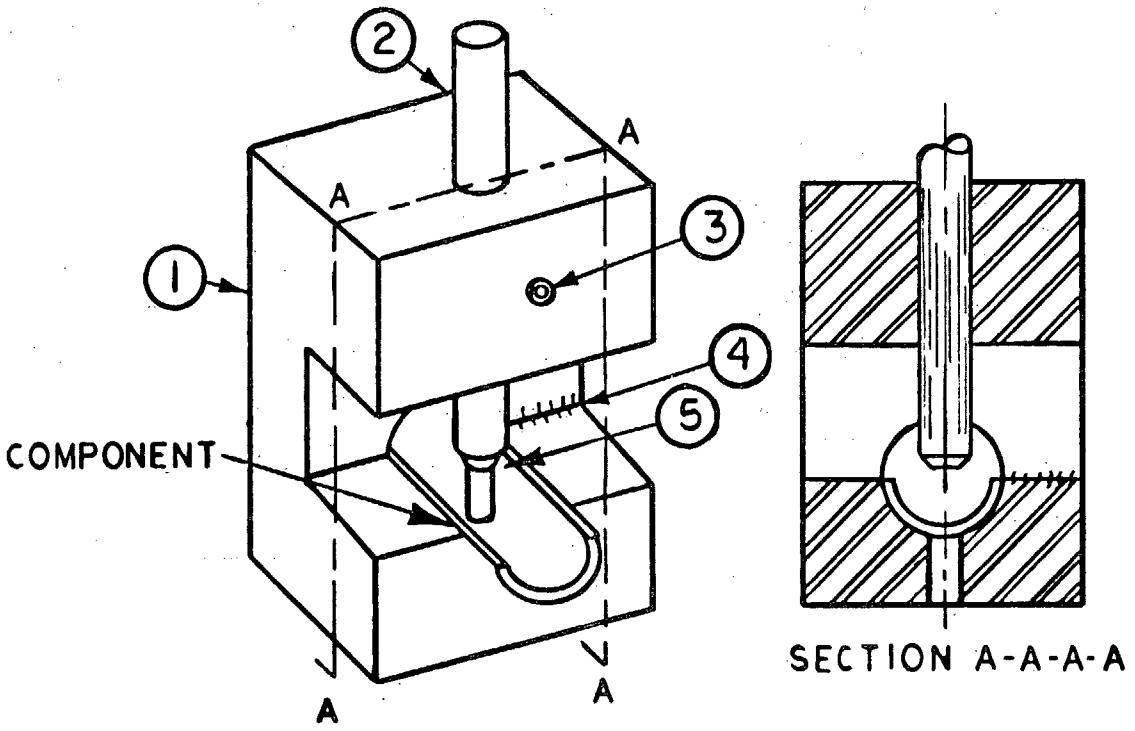
TEST NUMBER 1: NO BURST

Apply 57 pound compression load, remove body assembly. Item is rejected if evidence of bursting is observed. Tearing or puncturing of material shall be considered evidence of bursting.

TEST NUMBER 2: BURSTING

If the item is not rejected, replace it in fixture, using opposite end of body and apply 192 pounds compression load. If item shows no evidence of bursting it shall be rejected.

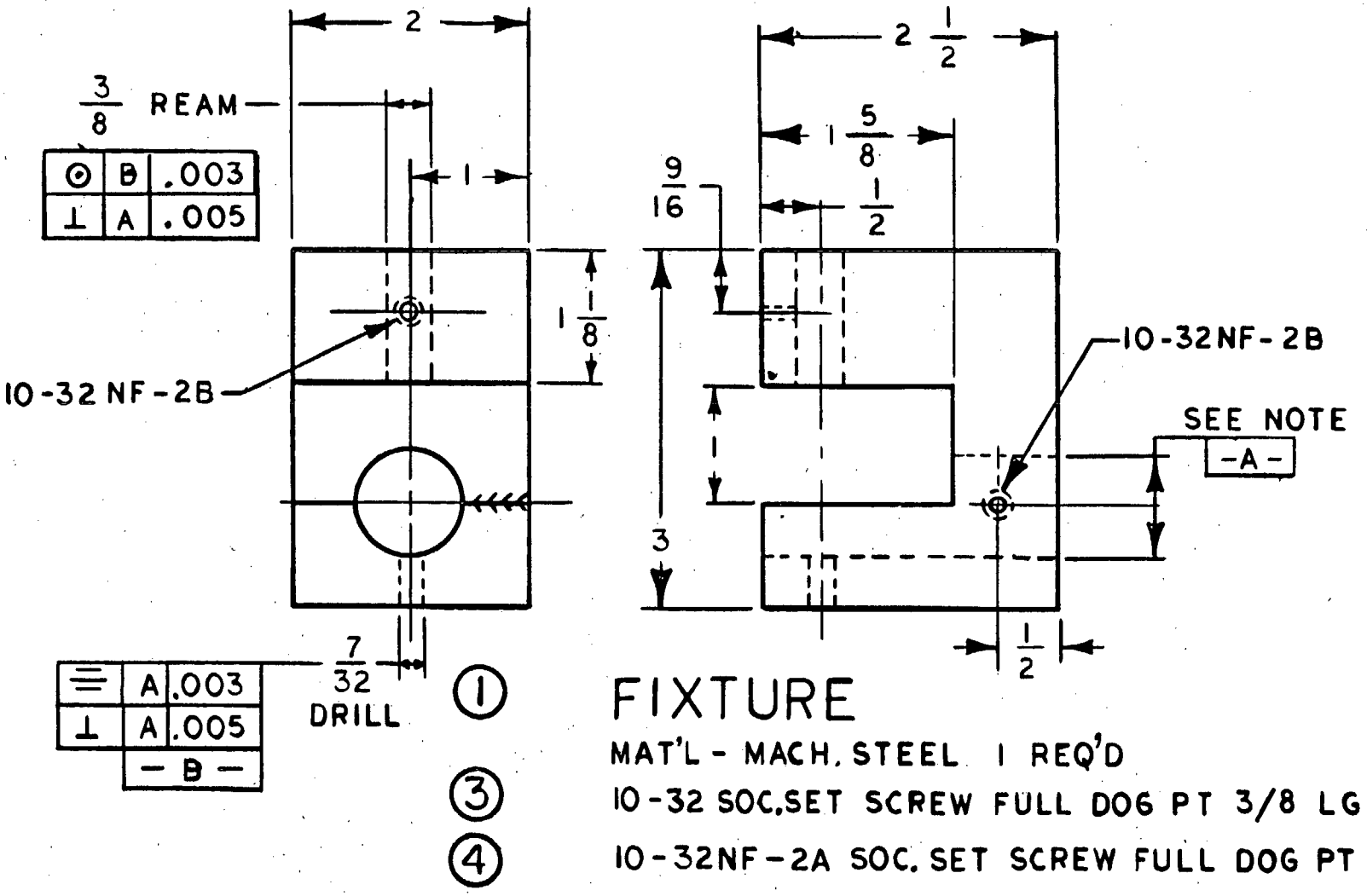
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ASSEMBLY
BUSTING STRENGTH
TEST FIXTURE

FIGURE 1 SHEET 1

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FIXTURE

MAT'L - MACH. STEEL 1 REQ'D

① 10-32 SOC.SET SCREW FULL DOG PT 3/8 LG

③ 10-32NF-2A SOC. SET SCREW FULL DOG PT 1/2 LG

④

NOTE DIAMETER TO SUIT APPLICABLE BODY ASS'Y

FIGURE 1 SHEET 2

DIMENSIONS ARE IN INCHES
TOLERANCES ON DICIMALS ±.005
FRACTIONS: 1/64
ANGLE ± 2°

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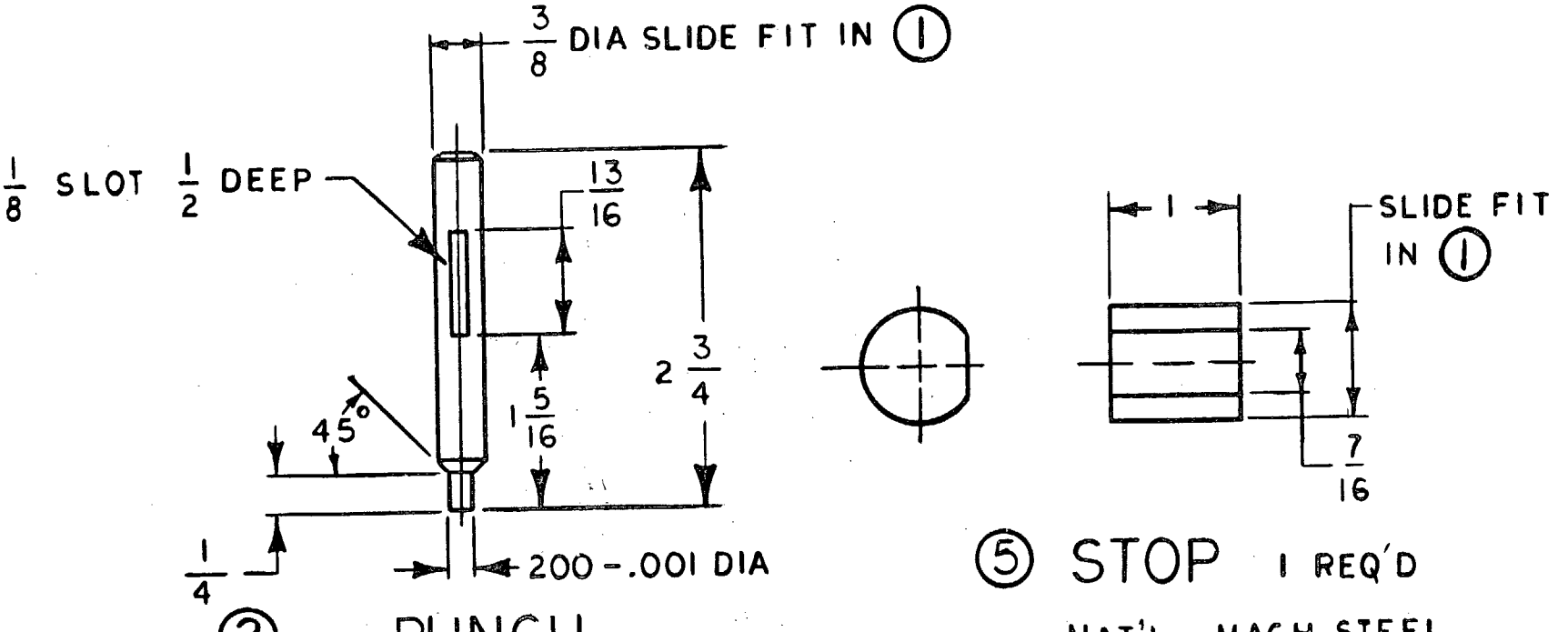


FIGURE 1 SHEET 3

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of Level C in MIL-STD-1169.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on Cartridge, Ignition, M752A1.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit equipment designs, as required, to Commander, U.S. Army Armament Research, Development and Engineering Center, ARDEC, ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

6.4 Visual examination qualifications. When compliance with the applicable requirements is in doubt as the result of visual examination, this characteristic may be measured or gaged to determine acceptability."

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A362)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48396A	CARTRIDGE, IGNITION, M752A1 PARTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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